ASSESSMENT REMIT

Name			
ID#			
Semester	Fall 2026	**Section**	
School	School of Engineering, Architecture and Interior Design (EAID)		
Program	Bachelor of Science In Computer Science/ Bachelor of Science in Information Technology		
Major	Computer Science/Information Technology		
Course Code	**CSCI 300**		
Course Title	Operating Systems		
Level	☐ Remedial ☐ Bridging ☑ Undergraduate ☐ Graduate		
Assessment Title	Operating System Resource Management Simulator		
Weighting	10 Marks		
Instructor	Dr. Maher Alrahhal		
Hand Out Date	27/10/2026		
Due Date & Time	Date: 01/12/2026	Time: 07:00 – 09:00 PM	
Feedback Post Date	08/12/2026		
Assessment Mode	☐ Individual ☑ Group ☐ Closed Book ☑ Open Book		
Assignment Length *(for assignment-based assessments)*	4–6 pages, excluding code and appendices		
Submission Mode	☐ Online by Email ☑ Online through LMS | TurnItIn ☐ Hard Copy		
Document Type	☐ Words ☑ PDF ☐ Artwork ☑ Software ☐ Other		

GENERAL REQUIREMENTS (ASSIGNMENT-BASED ASSESSMENT)

- The assignment must be completed in groups of **3 or 4 students only** (**maximum 4**). Groups of 2 or 5 students will not be accepted.
- All group members must contribute actively to the assignment.
- Students must use C programming to implement the required algorithms.
- Each group must submit a technical report (4–6 pages), source code, and screenshots showing the program results.
- The report must include group member names, Student IDs, and individual contributions.
- All source code must be tested and run successfully.
- Each group must demonstrate its program. Every student must participate in an **individual technical discussion** worth **3 marks** and explain their own contribution.
- Students must submit their assignment through the LMS before the specified deadline.
- All submitted work must comply with the University's Academic Integrity and Academic Misconduct Policy.

ASSESSMENT:

ASSESSMENT: OPERATING SYSTEM RESOURCE MANAGEMENT SIMULATOR

Instructions: Complete ALL FOUR (4) tasks. The assignment carries 10 marks, including 3 marks for individual technical discussion. Work in groups of 3 or 4 students and submit the C source code, results, and technical report.

Task 1 – Operating System Selection (2 Marks)

Situation: A university plans to establish a new computer laboratory for programming, web browsing, and office applications.

a) Compare Windows and Linux based on resource management, software compatibility, security, and cost.

b) **Recommend** the **operating system** for a given computing system based on **efficient resource management**. Justify your **recommend**ation for the university computer laboratory.

Task 2 – CPU Scheduling Simulator (3 Marks)

Situation: Five processes require CPU execution.

Process	**Arrival Time (ms)**	**Burst Time (ms)**
P1	0	8
P2	1	4
P3	2	2
P4	3	6
P5	4	3

a) Implement FCFS, non-preemptive SJF, and Round Robin scheduling (time quantum = 2 ms).

b) Calculate completion time, turnaround time, waiting time, and average waiting time for each algorithm.

c) Generate Gantt charts showing process execution.

d) **Select** the **suitable process scheduling algorithm** for a **given computing problem**. Use your calculated average waiting times to justify the **select**ion.

Task 3 – Memory Page Replacement (2 Marks)

Situation: A computer has three available memory frames.

Page Reference Sequence:

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2

a) Implement FIFO and LRU page replacement algorithms.

b) Show the frame contents after each page request.

c) Calculate the total page faults for both algorithms.

d) **Select** the **optimal paging technique** for a given computing system, **working in teams and communicating**. Compare FIFO and LRU results to justify the **select**ed technique.

e) Present your page replacement results, explain the selected algorithm, and include a group contribution table showing each member's responsibilities.

Task 4 – Individual Technical Discussion (**3 Marks**)

Each student must answer individually. The following three questions carry 1 mark each:

a) CLO1 (1 mark): Explain and justify your operating system recommendation from Task 1.

b) CLO3 (1 mark): Explain how your scheduling results support the algorithm selected in Task 2.

c) CLO4 (1 mark): Explain your optimal paging technique, page-fault evidence, and team contribution.

ANSWER GUIDELINES:

Students are expected to:

- Complete ALL FOUR assignment tasks, including **individual technical discussion** (**3 marks**).
- Use C programming to implement the required algorithms.
- Provide a comparison and justified operating system recommendation in Task 1.
- Implement and test FCFS, SJF, and Round Robin scheduling in Task 2.
- Show scheduling calculations and Gantt charts.
- Implement FIFO and LRU page replacement in Task 3.
- Present frame contents and page-fault calculations.
- Explain the results and justify algorithm selections.
- Submit a technical report, source code, screenshots, and a group contribution table. Every student must answer the **Task 4** discussion questions **individually**.

Presentation Requirements

- Include group member names and Student IDs.
- Submit a clearly structured technical report.
- Include relevant tables, figures, and screenshots.
- Ensure the source code runs correctly.
- Acknowledge external references and resources using APA style.
- Clearly describe each group member's contribution.

REFERENCING STYLE:

Students should use the American Psychological Association style (APA) as per the following guidelines: https://apastyle.apa.org/style-grammar-guidelines/references/examples

PLAGIARISM & GENERATIVE AI (FOR ASSIGNMENT BASED ASSESSMENTS):

Amity University Dubai is committed to a policy of honesty in all its operations, especially those pertaining to academic matters. It is your responsibility to use appropriate references and apply correct referencing practices. Unless explicitly stated otherwise, the use of generative AI to write an assignment is not permitted. Submission should be yours and should not be in breach of AMUD's Academic Integrity and Academic Misconduct Policy.

LATE SUBMISSIONS (FOR ASSIGNMENT BASED ASSESSMENTS):

You are advised to strictly adhere to the deadline. While Amity University Dubai might consider extenuating circumstances to support students who are experiencing unforeseen issues impacting their ability to complete assessments on time, any anticipated submission delay must be communicated in advance to the instructor who will consult with the Program Leader and the Dean before a permission is granted. All permissions must be based on valid reasons backed up by concrete evidence. For more details, you are advised to consult the latest version of AMUD's Academic Integrity and Academic Misconduct Policy.

FEEDBACK PLAN:

The preparation for the assignment will be supported through formative feedback during regular class sessions, laboratory activities, and office hours.
Students will receive feedback on their operating system comparison, scheduling implementation, page replacement algorithms, and technical report.
Summative written feedback will be provided on 08/12/2026.

ASSESSED COURSE LEARNING OUTCOMES:	RELATED QUESTIONS	MAX SCORE	MARK
CLO1: Recommend the operating system for a given computing system based on efficient resource management.	Task 1 + Task 4(a)	**3**	
CLO3: Select the suitable process scheduling algorithm for a given computing problem.	Task 2 + Task 4(b)	**4**	
CLO4: Select the optimal paging technique for a given computing system, working in teams and communicating.	Task 3 + Task 4(c)	3	
	TOTAL	**10**	

EMBEDDED PLO DIRECT ASSESSMENT QUESTIONS? ☑ **YES** ☐ **NO**

[IF YES, COMPLETE THE TABLE BELOW]

PLO	EMBEDDED QUESTIONS

PLO1: Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	Tasks 1–3 and individual discussion (Task 4)
PLO3: Communicate effectively in a variety of professional contexts.	Technical report, Task 3 explanation, and individual discussion (Task 4)
PLO5: Function effectively as a member or leader of a team, demonstrating leadership, collaboration and project management in achieving objectives related to the discipline.	Group collaboration, contribution evidence, and individual discussion (Task 4)

SCORING RUBRIC			**LEVEL OF ACHIEVEMENT**			
	DESCRIPTOR	**WEIGHT**	**4 [87%-100%]**	**3 [73%-86%]**	**2 [60%-72%]**	**1 [0%-59%]**
1	Operating System Selection (Task 1)	2	Accurate comparison and well-justified recommendation.	Good comparison with minor omissions.	Basic comparison with limited justification.	Incorrect or incomplete comparison.
2	CPU Scheduling Implementation (Task 2)	1	All three algorithms implemented correctly.	Mostly correct implementation with minor errors.	Partially working implementation.	Incorrect or missing implementation.
3	Scheduling Calculations and Evaluation (Task 2)	2	Accurate calculations, Gantt charts, and recommendation.	Mostly correct calculations and comparison.	Partially correct calculations or charts.	Incorrect or missing analysis.
4	Page Replacement Implementation (Task 3)	1	FIFO and LRU implemented correctly with accurate frame tracking.	Mostly correct implementation.	Partial implementation with errors.	Incorrect or missing implementation.
5	Page Replacement Evaluation (Task 3)	1	Accurate page-fault comparison and justified selection.	Good comparison with minor errors.	Limited comparison or justification.	Incorrect or missing evaluation.
6	Individual Technical Discussion (Task 4)	3	Independently explains the OS recommendation, scheduling selection, and paging selection accurately using results.	Explains all three selections with minor gaps or inaccuracies.	Partially explains the selections; limited evidence or understanding.	Cannot explain the work or provides incorrect answers.

Student Declaration

By signing this cover page, you acknowledge the instructions above (general requirements) and accept all responsibilities resulting from breaching them as per the related University policies. I hereby declare that it is my original work and I have duly acknowledged and referenced the source of any quotes, thoughts or ideas incorporated in the work.

No.	Student Name	Student ID	Student Signature	Date
1				
2				
3				

4				

To be filled by faculty

FEEDBACK FROM FACULTY:

Faculty Name and Signature **Date**